Contextual Driving Risks Analysis using Individualized Dynamic Smartphone-Based Data

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Abstract:
Traditional driving risk analysis reveals driving risk factors based on the correlation analysis at the aggregate level, and typically does not go down to the individual level due to the absence of personalized driving data. While some recent efforts applied the latest information and communications technology (ICT) for data collection and driving behavior evaluation, most researchers and practitioners currently rely solely on user GPS trajectories, without considering other critical risk factors in the surrounding environment that may also contribute to crash risk, i.e. the context-sensitive risk factors. A new approach that collects individualized driving behavior data from smartphone GPS module, combined with geographical network information and dynamic traffic conditions, is proposed to identify driving risk factors and evaluate driving behaviors in various contexts. The multi-source data reveals real world activity patterns as to when, where and how an individual driver performs, from which performance measurements on both the trip and driver level are defined to measure the risks in those patterns. Data is collected from a total of 69,283 trips from 255 users and numeric analysis is performed and summarized in this paper.

1. Introduction

Traditional driving risk analysis is usually performed based on crash history data or survey results, and uses this information to identify hazardous locations or driving risk factors contributing to the crash events. Examples of this kind of research include drinking and driving (Li et al. 2016); drug use and driving (Fergusson et al. 2008); gender differences (DeJoy 1992); age differences (Rhodes and Pivik 2011); years of driving experience (Jonah 1986); driver inattention (Tucker et al. 2015); personality and attitudes (Greaves and Ellison 2011), and so on. Research in this category reveals driving risk factors based on the correlation analysis at the aggregate level, and typically does not go down to the individual level due to the absence of personalized driving data.

An emerging stream of research has been applying the latest information and communications technology (ICT) to data collection and driving behavior evaluation in the last decade. Such data usually include not only vehicle spatial-temporal dimension location, but more importantly also speed, acceleration, and deceleration, which are frequently used to evaluate a driver's driving behavior. The 100-Car Naturalistic Driving Study (Neale et al. 2005) is the first instrumented vehicle study undertaken with the primary purpose of collecting large-scale naturalistic driving data. The resulting dataset includes approximately 2,000,000 vehicle miles, almost 43,000 hours of data, 241 primary and secondary drivers, 12 to 13 months of data collection for each vehicle, five channels of video and many vehicle state and kinematic variables. Similar data collection efforts can also be found in other studies. For example, 8 million second-by-second GPS data observations were collected from 106 drivers in Sydney over several weeks, and a framework and methodology for profiling drivers along multiple dimensions of behavior and risk to the driver and other road users was proposed (Ellison et al. 2012). Driving patterns of drivers who have (141 drivers) and have not (26 drivers) experienced crashes were investigated during a 14-month study period using the longitudinally collected GPS data in Atlanta (Jun 2006). Another study conducted an experimental field survey in Italy and UK collecting driving data from two instrumented vehicles with the goal of analyzing driving behavior in car-following conditions (Pariota et al. 2016).

While all of these efforts can be considered a major leap forward, one could argue that most researchers and practitioners currently rely solely on user GPS trajectories, which only measure the specific driver's performance, such as trip frequency, trip timing and hardness of braking, without considering other critical risk factors in the surrounding environment that may also contribute to crash risk, i.e. the context-sensitive risk factors. Examples of such risk factors include road conditions surrounding the vehicle of interest, location-related risk (e.g. an intersection with high risk) and dynamic traffic flow information. For example, if a driver exhibits stop-and-go or abrupt accelerate/decelerate behavior, without supplemental information on traffic conditions, it is usually difficult to tell if this is simply due to the driver's behavior or caused by heavy traffic conditions. Another intuitive idea is to compare the driver's instantaneous speed with the surrounding traffic. If a driver is observed to be consistently driving at a speed much higher than the other people around him/her, it's reasonable to assume that most likely this driver is driving more aggressively than average drivers.

In this paper, we propose a new approach that collects individualized driving behavior data from the smartphone GPS module, combined with geographical network information and dynamic traffic conditions, to identify driving risk factors and evaluate driving behaviors under various contexts. The multi-source data collected in this research reveals real world activity patterns as to when, where and how an individual driver performs, and performance measurements for both trip- and individual-level are defined to measure the risks in those patterns. In addition, the crash history of the drivers is also collected through a designed survey, so the relationship between the defined driving performance measurements and accident rate can be examined and verified. After data collection and performance measurement definition, multiple data mining techniques are applied to analyze the context-sensitive driving behavior based on a total of 69,283 trips from 255 users collected between January 2015 and January 2016.

2. Data

2.1. Travel-Related Data

The primary travel-related data includes three main categories: vehicle trajectory data, roadway geometry information, and time-varying traffic dynamic condition. Individualized GPS trajectory data is collected through a smartphone app "Metropia" to keep track of people's driving behavior, and the system cloud server also stores the information on surrounding traffic and geometrical network condition. More details on the data collection methods are explained in Zhu et al. (2013).

Vehicle trajectory data

Vehicle trajectory data is used to extract driving performance such as acceleration, driving speed, instant location. When a user starts a trip with the app, the internal GPS module built in the smartphone is activated and starts to record the second-by-second data. These data, including detailed position such as latitude, longitude, heading, timestamp, velocity and corresponding link in the roadway network will be collected at fine time interval and sent back to the cloud server, where they will be stored and used for further analysis. A total of 69,283 trips were collected for this research.

Roadway geometry information

The geographic network dataset is stored on the cloud server and includes a set of links and nodes, i.e. $G = (N, A)$ where N is the set of nodes and A the set of links. The key attributes relevant with this research include location of the intersections and links, type of a road segment (e.g. freeway, highway, arterial, etc.) , speed limit, link length, turn connections between links (e.g. left turn, right turn, etc.) and so on. This network data is originally used for routing purposes, and can be joined with the traffic information fairly easily, and linked with GPS trajectory data through a map matching module.

Time-varying traffic dynamic condition

Time-varying traffic dynamic condition, including historical traffic dataset and real time traffic information, is also stored on the server, originally for routing purposes. Such data describes dynamic traffic condition information with 5 minutes as its interval. For analysis purposes, the three types of dataset in this Section 2.1 are joined together based on index of links before it's further processed and analyzed.

2.2. Crash-Related Data

From a pool of frequent users (defined as those who had completed at least 10 trips with the smartphone app), we sent out surveys to collect information about their traffic accident history in the past and demographic information. In total, 260 users responded to this survey, among which 116 were from Tucson AZ, 142 were from Austin TX, and 2 were from New York. After excluding 5 users without good quality GPS data or from

New York where the sample size is small, 255 users with both travel related data and crash related data are studied in this paper.

The drivers' ages range from 19 or younger to over 65 years old, with an average age at 37 years of old. There are 136 female users and 119 male users in the sample. Among them, 243 (around 95%) of the drivers have been driving for more than 5 years. The percentage of people driving over 10,000 miles annually is 60%. The average VMT (vehicle mileage traveled) per year is about 10,800 miles.

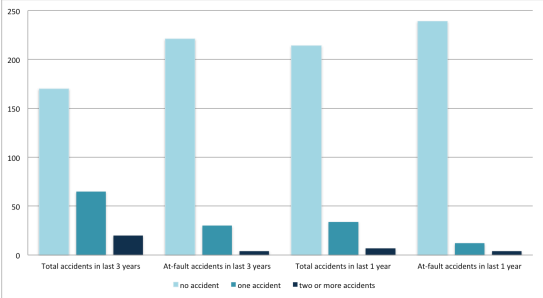


Figure 1 Number of users with at-fault or total accidents in last one and last three years

In our model, we will focus on the accidents in the recent years, (i.e. within the past one year and three years). Out of 255 users, 209 (about 80%) reported that they have at least one accident in the past years and 46 reported having no accident at all in the past. Forty-one users reported having at least one accident within the last one year, and 28 and 35 reported having an accident 1-2 years ago and 2-3 years ago, respectively. Figure 1 shows the number of people with accidents in the past three years. When considering the total number of accidents, there are 214 users without any accident in the last one year, and 170 users with no accidents in the last 3 years. Additionally, at-fault accidents will be examined separately from all accidents, as we expect different impacts of driving performance on risks for all accidents versus at-fault accidents. From Figure 1, 239 users reported no at-fault accident in the last year, 12 users reported one at-fault accident and 4 reported more than one accident. These numbers change to 221, 30, and 4, respectively by looking back for the last 3 years for at-fault accidents.

3. Contextual Driving Performance Measurement

The performance measurements are based on the travel-related data as described in Section 2.1. As shown in Figure 2, five measurements are defined to capture the driving performance at the trajectory and link level. Based on that, the trip level performance is then aggregated for each trip from the trajectory and link level measures. For the drivers who have completed more than ten trips, the driver level performance is further aggregated from all trips completed by this driver. The following sections describe the details of each step.

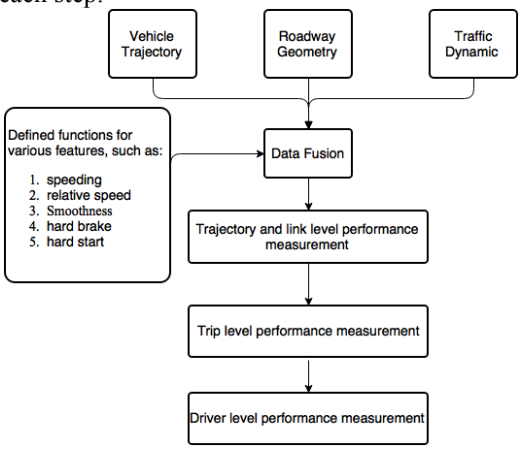


Figure 2. Flow chart of driving performance measurement calculation

3.1. Trajectory and link level measurements

In total, five features are defined for driving performance at this level, including relative speed, speeding, smoothness, hard brake and hard start. The two contextual features, relative speed and speeding, are obtained from the instant driving performance and context for that specific instance. The other three features, smoothness, hard brake and hard start, are purely from instant driving performance. In this section, we describe how we define each measure.

The **relative speed** measures the variations between a driver's driving speed and the real time traffic speed on the same road segment at the same time. The pure variations in instantaneous speed and traffic speed at link l is calculated as $r_l = V_l - \bar{v}_l$,

V_l is the user's driving speed at link l calculated from GPS speed.

$\bar{v}_l$ is the average link speed from Traffic Dynamic data stored on the server, which is obtained from a national data provider and is originally used for routing purposes.

If r_l is positive, it means that at this link, this driver is driving faster than others on average, and vice versa. A higher absolute relative speed value stands for a larger discrepancy in the driving behaviors between this driver and other drivers on

the same link, i.e. this driver either drives much faster, or much slower compared with other drivers, which indicates unsafe driving behavior. If the variation is high, it indicates that the driver does not follow the traffic consistently and has a higher tendency to bypass other vehicles. If the variation is low, it means that this driver is very conservative in driving and can be a bottleneck in the traffic, and other vehicles need to change lanes to pass it. The larger absolute value of r_l represents a higher risk.

If we consider the driving context, the same variation r_l may have a different risky impact at slow versus fast traffic flow conditions, or heavy versus light traffic conditions. For example, if someone drives 7mph faster than the others on an arterial with 45 mph average speed, he may just encounter free flowing traffic condition. But if it is at a heavily congested road with average speed at 15mph, he may need to keep changing lanes to pass other vehicles, which is more dangerous. Similarly, driving 7mph faster than the traffic indicates risk if everyone else is driving at 40mph on average, but the risk might be different if he is driving on the freeway with the same 7mph deviation while the average traffic speed is 60mph. The contextual feature is designed by involving traffic condition simultaneously with their speed variations r_l. Therefore, the contextual relative speed risk R_l is constructed as

$$R_l = F(r_l, \bar{v}_l, c_l)$$

The value of R_l is direct proportional to $|r_l|$, and $\bar{v}_l$ and inverse proportional to c_l. The same R_l is assigned to each trajectory on this link l, and $R_t = R_l$ if the vehicle is located at link l at timestamp t.

c_l is defined as congestion level, calculated from $\bar{v}_l$ and free flow speed FFS_l of link l. $c_l \in [0,1]$, indicating free flow condition while $c_l = 1$ and complete congestion while $c_l = 0$

Speeding describes the difference between user's driving speed and posted speed limit. By comparing the speed limit and vehicle's instantaneous speed, the pure speeding performance s_t at timestamp t can be calculated as

$$s_t = V_t - SL_t$$

V_t is user's driving speed at timestamp t from the vehicle GPS data

SL_t represents the speed limit at timestamp t, obtained from corresponding speed limit of a link SL_l from the Roadway Geometry data, matched by link index l the vehicle is located at time t.

This speeding performance should be adjusted with contextual status, such as type of road and/or congestion levels. For example, 10mph over speed limit may impose lower risks on freeway than on arterials. Similarly, the risk imposed by speeding may vary between congestion and free flow conditions. Thus, a contextual speeding performance measure that captures the both the speeding and traffic conditions can be written as follows:

$$S_t = G(s_t, c_t)$$

where c_t is congestion level at timestamp t, equal to c_l, matched by link index l where the vehicle is located at time t.

Smoothness measures the stability in acceleration and deceleration behavior. $sm_t = 1$, if $|\alpha_t| < threshold_\alpha$ and $V_t \geq 10\ mph$, where $\alpha_t = (V_{t+1} - V_t)/(t_{t+1} - t_t)$ is the acceleration at timestamp t. This threshold $threshold_\alpha$ is used to capture minor or zero acceleration and deceleration, and typically the value is within the range 0.5 and 1. When a vehicle is in motion (>10mph), a higher value indicates that the acceleration and deceleration of this vehicle is smoother, and that this driver has better control over the vehicle to remain driving at certain constant speed.

Hard brake and **hard start** measures the acceleration and deceleration process in the vehicle operation. In general they are low-probability events as less than 10% of the trips have at least one hard brake or hard start in a trip. Hard brake flag $HB_t = 1$, if $\alpha_t < threshold_{HB} < 0$, and hard start flag $HS_t = 1$, if $\alpha_t > threshold_{HS} > 0$. These two measures are normalized to the frequency of hard brakes and hard starts by dividing the count of events by the duration of the trip. The thresholds value $threshold_{HB}$ is typically chosen from a value lower than -12 ft/s^2, and the value used as $threshold_{HS}$ is typically larger than 10 ft/s^2. Similar to the **Smoothness** measure, there are no contextual features included in these variables either.

3.2. Trip level measurements

The trip level performance is aggregated from the measurements P_t at trajectory level within this trip, which can be considered as weighted average with time duration D_t as the weight value. P_t can represent all the five trajectory level measurements, including R_t, S_t, sm_t, HB_t, HS_t. D_t is the duration from timestamp t to $t+1$. P^i is the trip level performance measurement for trip i, which is

firstly calculated as P^i, and then normalized by function $g()$ among the set of all selected trips I. The normalized value indicates the performance evaluation of this trip relative to all of the trips, with 0 being the worst performance and 1 being the best performance among all the trips

$$P^i = \frac{\sum_t (D_t * P_t)}{\sum_t D_t}$$

$$P^{i*} = g\left(P^i | \{P^{i'}, i' \in I\}\right)$$

3.3. Driver level measurements

In order to evaluate the driving performance for each driver, the performance measurements from each trip are further aggregated to the driver level, with extracted features describing the average performance and worst-case performance, respectively. The description of each variable can be found in Table 1. The first five variables, ending as "_m" are the weighted average of the trip level measures for each individual, and the next five variables ending as "_w" are the worst case measures among all the trips of this individual. The higher the value, closer to 1, indicates that the driver performs better for certain measures on average, or in the worst case scenario. For example, the minimum, mean and maximum values of rel_m, the average relative speed measure for a driver, indicate that among all the drivers, their average performance calculated from their trips, range from 0.41 to 0.68. And a driver with a "rel_m" value greater than 0.56 is performing better than the average relative speed performance of all the drivers.

Table 1. Summarized description for driver-level driving performance variables

Variables	Definition
rel_m	Average relative speed measures weighted on duration of each trip among all the trips for one user
spd_m	Average speeding measures weighted on duration of each trip among all the trips for one user
smooth_m	Average smooth measures weighted on duration of each trip among all the trips for one user
HB_m	Average hard brake measures weighted on duration of each trip among all the trips for one user
HS_m	Average hard start measures weighted on duration of each trip among all the trips for one user
rel_w	Worst relative speed measures among all the trips for one user
spd_w	Worst speeding measures among all the trips for one user
smooth_w	Worst smooth measures among all the trips for one user
HB_w	Worst hard brake measures among all the trips for one user
HS_w	Worst hard start measures among all the trips for one user

The additional seven variables measuring traffic and roadway conditions are also aggregated using the weighted average to indicate the context that each driver frequently experienced. The description of each variable can be found in Table 2.

Table 2. Summarized description for driver-level contextual environment variables

Variables	Definition
Congestion_m	Average congestion level among all the trips for one user
Freeway_m	Average percentile duration on freeway among all the trips for one user
Arterial_m	Average percentile duration on arterial among all the trips for one user
Local_m	Average percentile duration on local street among all the trips for one user
Peak_m	Average percentile duration at peak hour among all the trips for one user
Offpeak_m	Average percentile duration at off peak hour among all the trips for one user
Night_m	Average percentile duration at midnight among all the trips for one user

4. Driving Performance and driving risks Analysis

In this paper, multiple data mining techniques are applied to analyze the context-sensitive driving behavior based on the massive amount of trajectory data collected between January 2015 and January 2016 for the 255 users.

Specifically, efforts are made to firstly identify the most important performance measurements to characterize an individual's driving pattern at the driver level, and to quantify the causality relationship between various driving performance measurements and the drivers' accident risks. In this section we present our empirical analysis on the relationship between driving performance and accidents using driver level data.

4.1. Driver-level clustering analysis among driving performance measurement variables

Given the potential high correlations among some of the risk measurements defined earlier, principal component analysis (PCA) is performed among the variables in Table 1 to reduce the number of variables and redundancy in the subsequent modeling process.

As revealed in Figure 3, among the 10 driving performance variables, about 80% of the variance is captured by the first component (variances=0.16 at Comp 1) and the second component (variances=0.05 at Comp 2), while Comp 3 and all others only capture a total variance of 0.06, which is about 20% out of the total variances. Thus, we fit the models with two components. The results of the PCA analysis are shown in Table 3.

From the results, we can see the first principal (PC1) is mainly contributed by the information in the following four variables: HB_m HS_m, HB_w HS_w. These variables reveal the actions on the pedal (braking pedal and gas pedal) when operating the vehicle. Intuitively, smooth pedal operation is not only safer driving practice, but also good to improve fuel efficiency and reduce vehicle emissions.

The second principal component (PC2) captures the information in rel_m, spd_m, rel_w, spd_w, and smooth_w. These variables can be identified as vehicle speed related operation performance. Drivers with higher relative speed either tend to speed, or exhibit frequent lane changing behavior, introducing more risks in the driving behavior. On the other hand, good car following behavior will be beneficial to reduce driving risks, particularly for heavy traffic conditions. Intuitively, a vehicle with higher deviation in its driving status from the others, either driving much faster than the other drivers, or much slower than average traffic flow, will bring disruptions to the traffic flow, which will not only lower system efficiency, but also force other drivers to take certain actions such as deceleration or acceleration – neither one is considered a safe driving behavior.

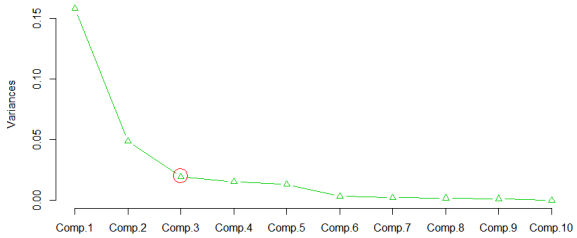


Figure 3. Variance captured in principal component analysis

Table 3. Results table of PCA analysis

Standardized loadings (pattern matrix) based upon correlation matrix					
	PC1	PC2	h2	u2	com
rel_m	-0.22	0.58	0.387	0.61	1.3
spd_m	-0.03	0.55	0.302	0.7	1
smooth_m	-0.18	0.14	0.051	0.95	1.9
HB_m	0.89	-0.04	0.788	0.21	1
HS_m	0.89	-0.08	0.804	0.2	1
rel_w	0.09	0.78	0.616	0.38	1
spd_w	0.2	0.82	0.706	0.29	1.1
smooth_w	0.19	0.54	0.324	0.68	1.2
HB_w	0.77	0.34	0.707	0.29	1.4
HS_w	0.76	0.31	0.672	0.33	1.3

	PC1	PC2
SS loadings	2.91	2.44
Proportion Var	0.29	0.24
Cumulative Var	0.29	0.54
Proportion Explained	0.54	0.46
Cumulative Proportion	0.54	1

Mean item complexity = 1.2

Test of the hypothesis that 2 components are sufficient.

The root mean square of the residuals (RMSR) is 0.14

with the empirical chi square 444.16 with prob < 1.10E-77

Fit based upon off diagonal values = 0.82

Based on the PCA results, we reduce the driving performance to two principal factors: pedal (PC1) and speed (PC2). These two factors can be used as explanatory variables in regression analysis (discussed in Section 4.2). Additionally, using these two factors, we also construct a categorical explanatory variable from the clustering method (k-means method) to categorize drivers into five groups with similar driving behaviors, as shown in *Figure 4*. Each driver is categorized to one of the five groups identified. Users in the first group G1 (size=51) have good performance in terms of both pedal and speed operation. The group G2 (size=88) is also performing well with pedal operation, but not as well as G1 in speed control; which indicates they may have been speeding or have had a larger deviation from the average traffic speed. The next group, G3 (size=62), includes drivers with frequent speeding or inconsistent speed behavior, but good pedal operation. G4 (size=30) stands for the portion of drivers with good performance in vehicle speed but not very good in pedal operation. The last group, G5 (size=24), includes drivers whose performance is not good in both pedal and speed operation, with very frequent hard operations on either brake or gas pedal, and travel speeds severely deviated from traffic flow or speed limits.

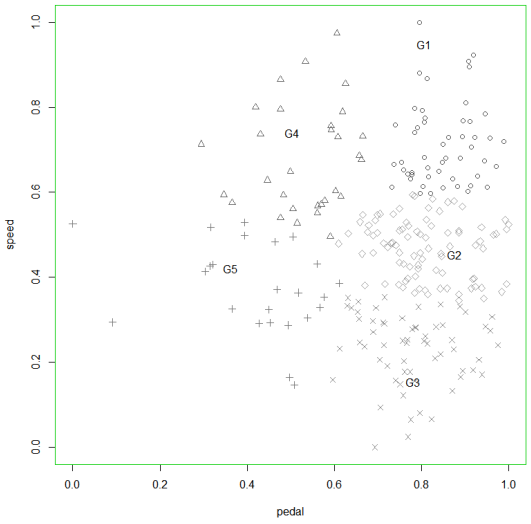


Figure 4. Category of drivers from the cluster result of speeding and pedal behavior

4.2. Risk performance analysis

With the data on driving behavior (discussed in Section 3) and accident history (discussed in Section 2.2) for the same group of users, we are able to examine the relationship between driving performance and accident rate. The purpose of the analysis is to use the crash data to validate the idea that drivers with good driving performance are less likely to run into vehicle accidents, or are those who are driving safely and conservatively to avoid traffic accidents. In contrast, those who do not perform as well on driving are more likely to get involved in auto accidents. The linkage of the current driving habits with accident history assumes that the current driving behavior represents the habitual behavior in the past few years, and driving behavior has not changed significantly in the past few years.

The studied response variable, number of accidents, is in the format of count data, and can also be constructed as a binary categorical variable or an ordered response variable. Therefore, the potential statistical models may include several forms of count data models, or binary logit model or ordered probit model Poisson and negative binomial based models are the most widely-used methods in crash-frequency analysis as extensively reviewed in Lord and Mannering (2010), and Mannering and Bhat (2014). These methods are mostly used to address the problem as a number of accidents during a time period in geographical locations (such as roadway segments), including applications of random effect models (Guo et al. 2010), random-parameter models (Yu et al. 2015), and mixture models (Park et al. 2014).

The frequency of the accidents can be converted to the time between crashes, and thus duration models, such as the hazard-based model, model the factors affecting the time duration between their crashes over their lifetimes (Mannering 1993). Studies focused on individual drivers are limited in literature because the response timing for crashes is hard to achieve and often relies heavily on self-reporting. This method may not be applicable for this research, because of the disadvantages in the precision of the crash timing collected in the survey.

The number of accidents can also be analyzed as an order discrete response (Greene and Hensher 2010), or other complex methods such as generalized ordered-response models in Castro et al. (2012), or Generalized Extreme Value (GEV) models discussed by Paleti (Paleti 2016). Also, the model can be constructed as a binary logit model for the response variable as whether or not the drivers have experienced an accident in the past few years. There are more methods discussed in Lord and Mannering (2010) and Mannering and Bhat (2014).

We start the analysis by modeling the data with the negative binomial regression, which is easy to estimate, and accounts for overdispersion because of the following two reasons:

- The focus of this research is quite different from what has been discussed in the previous studies. The driving performance measures, such as vehicle speed and pedal maneuvering in the Section 3 and Section 4.1, are an emerging type of data, similar to the naturalistic driving data. The explanatory variables in this research used to be considered the unobserved heterogeneity portion in the previous modeling approaches, and have not been analyzed and measured in quantity.
- The response variable selected in the final model is the number of at-fault accidents in three years. The distribution of the response variables is shown as the second block of bars in Figure 1. The overdispersion pattern can be implicated from the larger variance value (0.3) compared to the mean value (0.17).

The results from three negative binomial models are shown in Table 4. All these models use the number of at-fault accidents in the past three years as the response variable, and also age and vehicle mileage travelled reported by each driver

as the explanatory variables. The differences exist in the driving performance measures used in each model. Model 1 uses the detailed driver-level variables from Table 1. Model 2 uses the principal component variables achieved from Table 3. And Model 3 uses the category of the drivers defined from the clustering model shown as Figure 4. The sample size used in each model is 255, for the users who ever responded to our survey.

Table 4 Results of negative binomial regression models for the impact of driving performance on the number of accidents in the past three years

Variables	Model 1			Model 2			Model 3		
	Estimate	z-value	sig	Estimate	z-value	sig	Estimate	z-value	sig
Constant	3.99	1.652	.	-1.84	-8.327	***	-2.38	-6.610	***
Age (35 to 39)	-2.45	-2.271	*	-2.36	-2.205	*	-2.38	-2.249	*
Age (45 to 54)	-1.17	-1.872	.	-1.16	-1.814	.	-1.18	-1.885	.
VMT (more than 15k miles)	1.40	3.679	***	1.25	3.064	**	1.29	3.295	***
Average smooth pedal performance	-7.42	-2.164	*						
Worst relative speed performance	-3.96	-2.652	**						
Principal component (speed maneuver)				-0.35	-1.817	.			
Driver category G2							0.84	1.893	.
Driver category G3							0.90	1.949	.
Null deviance	156.37	Df=	254	139.87	Df=	254	147.77	Df=	254
Residual deviance	115.77	Df=	249	111.88	Df=	250	116.56	Df=	249
AIC	225.16			232.33			232.57		
R-square	0.26			0.20			0.21		
McFadden R-square	0.14			0.10			0.11		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

As can be seen from Table 4, in all models VMT is positively significant, which suggests that people who drive more are more likely to have at fault accidents. Additionally, drivers in the age group of 35 to 39, and 45 to 54 are found to have fewer at fault accidents. Driving performance measures defined in Section 3.2 are also found to demonstrate significant relationship with number of accidents. For each of the five performance measures (relative speed, speeding, smooth, hard brake and hard start), we calculated both the average and the worst performance values for each driver. Among the 10 different measures, our Model 1 results show that drivers with better performance in the smoothness of acceleration and deceleration on average and drivers whose worst-case performance in relative speed (car following) maneuverer is better than other drivers have significantly fewer accidents than others. These findings suggest that a driver with good habitual driving performance is less likely to have accidents or to be at fault in vehicle accidents, which is consistent with our expectation. The other variables from Table 1 are not statistically significant in our model. One possible explanation is that these variables may have certain unobserved correlation with the average smooth performance or worst-case relative speed performance, and their relationships with accident risks are captured by the two significant variables.

Model 2 confirms again the connection between the driving performance and driving risks. The second principal component (PC2) in Table 3, mainly projected from relative speed and speeding variables, is significantly negative. The other principal component that represents pedal action is not significant in the model result. Our findings suggest that speed control maneuverers have a stronger influence on the risks of at-fault accidents than the pedal maneuverers.

This finding is further validated in Model 3. As illustrated in Model 3, drivers in G2 and G3 are found to have more at fault accidents. Recall drivers in G2 and G3 do not have good speed control, meaning they may have been speeding or tend to drive at a speed that deviate from others. The findings that drivers in G2 and G3 have more

at-fault accidents than drivers in G1 further confirm that speed related driving behavior is relevant to the chance of having an auto accident. Overall, the relationship between the driving performance and risks of at-fault accidents are verified in all models.

5. Conclusions

In this paper, we propose a new approach that collects individualized driving behavior data from the smartphone GPS module, combined with geographical network information and dynamic traffic condition, to identify driving risk factors and evaluate driving behaviors under various contexts. Performance measurements for both trip-level and individual-level are defined in this paper to measure the risks in those patterns. In addition, crash history of the drivers is also collected through a designed survey, so the relationship between the defined driving performance measurements and accident rate can be examined and verified.

Through the numeric analysis, principal component analysis is firstly performed which reveals two main components, namely pedal and speed can capture the majority of the driving behavior. Based on these two components, clustering methods are applied to divide the drivers to five groups. The subsequent regression analysis on crash history and driving performances verifies both pedal operation and driving speed are significantly related to more at fault accidents, which validates the modeling and analysis efforts in this study. Among all the driver level variables, drivers with the better average smooth acceleration/deceleration, and better performances of relative speed in their worst-case are less likely to be involved in at fault accidents.

Our unique contribution lies in the contextual-sensitive performance measurements in relative speed, speeding, and smoothness, derived from the vehicle speed, traffic condition, and roadway information jointly. The study results reveal the benefits of integrating multi-source spatial-temporal information in the driving risk analysis process. In addition, due to the other advantages of the smartphone-based GPS data collection, including lower hardware cost compared to onboard telematics devices, and the capability of collecting data from a greater population, this approach has the great potential of further broadening the current adoption of UBI programs.

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